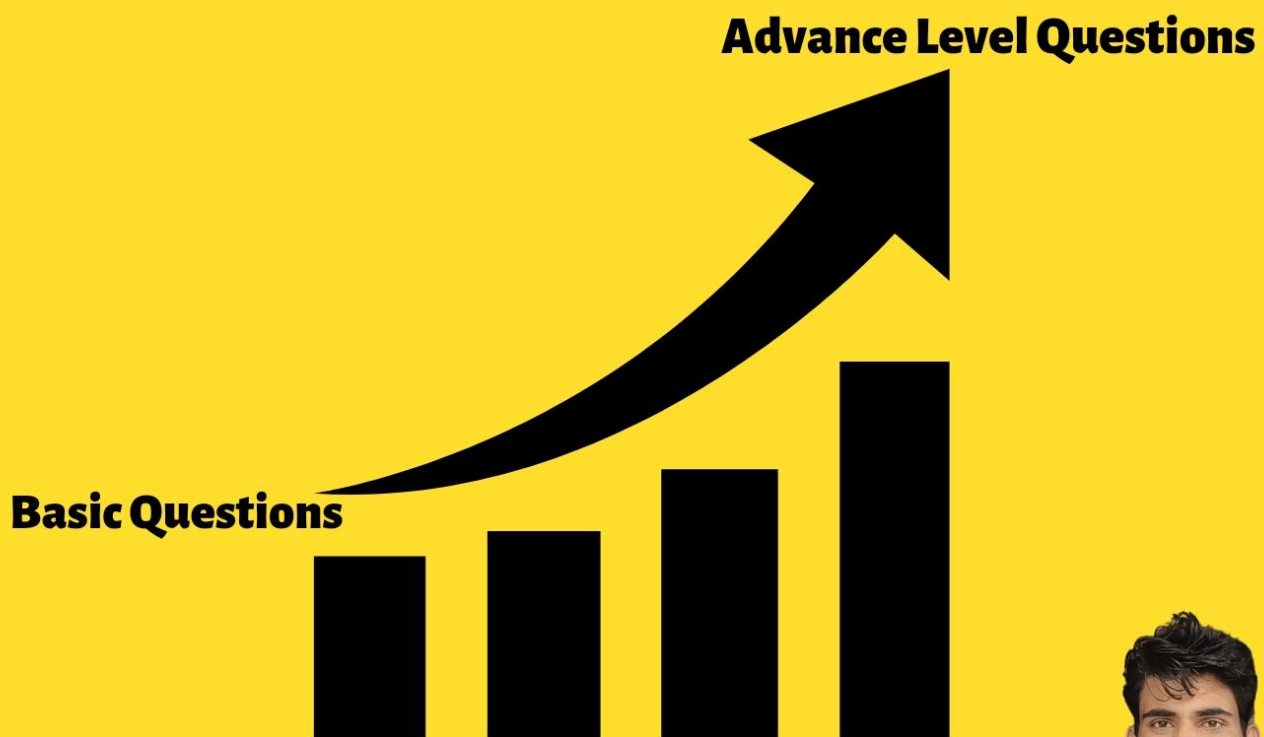


MATHS

AVERAGE

ALL TYPE WISE QUESTIONS



Maths with Gajanand



Average

$$* \quad \text{Average} = \frac{\text{Sum of Quantities}}{\text{Numbers of Quantities}}$$

Ex:- 6, 12, 15, 17, 25 avg = ?

Soln

$$\text{avg} = \frac{(6+12+15+17+25)}{5} = \frac{75}{5} = \boxed{15}$$

m-II Deviation method :-

$$\begin{array}{ccccccccc} 6 & 12 & 15 & 17 & 25 & & & & \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & & & & \\ -3 & -3 & 0 & +2 & +10 & & & & \Rightarrow (-12+12=0) \end{array}$$

Total net deviation = 0

Ex:- 40, 70, 50, 80, 110 avg = ?

$$\begin{array}{ccccc} \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ -10 & +20 & 0 & +30 & +60 \end{array}$$

(Let avg = 50)

$$\text{net deviation} = (-10+20+30+60) \Rightarrow 100$$

$$\text{avg} = 50 + \frac{100}{5}$$

$$= 50 + 20 = \boxed{70} \text{ Ans}$$

average of Series

1. Sum of 1st n Natural no. = $\frac{n(n+1)}{2}$

$$\text{average} = \frac{\frac{n(n+1)}{2}}{n} = \boxed{\frac{n+1}{2}}$$

2. $1^2 + 2^2 + 3^2 + 4^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$

$$\text{average} = \frac{(n+1)(2n+1)}{6}$$

3. $1^3 + 2^3 + 3^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$

$$\text{average} = n \left[\frac{(n+1)}{2} \right]^2$$

4. Sum of 1st n even no. = $n(n+1)$

$$\text{average} = (n+1)$$

5. Average of 1st n odd no. = $\frac{1+(2n-1)}{2} \leftarrow \left(\frac{\text{first} + \text{Last}}{2} \right)$

$$1, 3, 5, 7, \dots, (2n-1) = \boxed{n}$$

⇒ Consecutive no. / Arithmetic progression:-

↪ average = middle number

Ex:-

11, 12, 13, 14, 15, 16, 17

↑↑

Avg.

OR:-
$$\text{Avg} = \frac{\text{first} + \text{Last term}}{2} \Rightarrow \frac{11+17}{2} = \boxed{14}$$

Ex:

4, 5, 6, 7, 8, 9 avg = ?

avg = $\frac{6+7}{2} = 6.5$

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Ex:-

$\underbrace{24, 25, 26, 27}_{\downarrow} \quad \underbrace{28, 29, 30}$

$\text{avg.} = 27 \Rightarrow \frac{24+30}{2} = \frac{25+29}{2} = \frac{26+28}{2}$

Queⁿ:- Average of the 1st 100 positive integers is:

SOP:

1, 2, 3, - - - - - 100

$$\text{avg.} = \frac{1+100}{2} = \boxed{50.5} \text{ Ans}$$

1. What is the average of the squares of the first 19 natural numbers?

Soln:-
$$\text{avg} = \frac{(n+1)(2n+1)}{6} \Rightarrow \frac{20 \times 39}{6} = \boxed{130} \text{ Ans}$$

2. Find the average of cubes of first 49 positive integers.

Soln:-
$$\text{avg} = \frac{n(n+1)^2}{4} \Rightarrow \frac{49 \times (50)^2}{4} = \boxed{30625} \text{ Ans}$$

3. What is the average of first 10 even numbers?

Soln:-
$$\text{avg} = (n+1) \Rightarrow \boxed{11} \text{ Ans}$$

4. The average of all the odd integers between 2 and 22 is?

Soln:- 3, 5, 7, ..., 21

$$\text{avg} = \frac{\text{first} + \text{Last}}{2} = \frac{3+21}{2} = \boxed{12} \text{ Ans}$$

5. Average of all even numbers between 222 and 250 is:

Soln:- 224, 226, ..., 248

$$\text{avg} = \frac{224+248}{2} = \boxed{236}$$

6. The arithmetic mean (average) of the first 10 whole number is:

Soln:- 0, 1, 2, ..., 9

$$\text{avg} = \frac{0+9}{2} = \boxed{4.5}$$

[7] The average of prime numbers between 1 and 20 is -

Soln:- Prime No. $\rightarrow$ 2, 3, 5, 7, 11, 13, 17, 19

$$\text{avg} = \frac{(2+3+5+7+11+13+17+19)}{8} = \boxed{\frac{77}{8}} \text{ Ans}$$

[8] What is the average of first 7 multiples of 7?

Soln:- 7, 14, --- 49 $\text{avg} = \frac{7+49}{2} = \frac{56}{2} = \boxed{28}$

[9] What is the average of all numbers between 9 and 90 which are divisible by 8?

Soln:- 16, 24, --- 88 $\text{avg} = \frac{16+88}{2} = \boxed{52} \text{ Ans}$

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[10] What is the average of all numbers between 8 and 74 which are divisible by 7?

Soln:- 14, 21, --- 70 $\text{avg} = \frac{14+70}{2} = \boxed{42} \text{ Ans}$

[11] What is the average of the first six (Positive) odd number each of which is divisible by 7?

Soln:- 7, 21, 35, 49, 63, 77 (same difference = +14)

$$\text{avg} = \frac{7+77}{2} = \boxed{42} \text{ Ans}$$

[12] a, b, c, d, e, f, g are consecutive even numbers, j, k, l, m, n are consecutive odd numbers. The average of all the numbers is:

Soln:- (a) $3\left(\frac{a+n}{2}\right)$ (b) $\left(\frac{l+d}{2}\right)$ (c) $\frac{a+b+m+n}{4}$ (d) $\frac{j+c+n+g}{4}$

$a, b, c, d, e, f, g \rightarrow \text{avg} = d$ (middle term)
 $\rightarrow \text{sum} = 7d$

$j, k, l, m, n \rightarrow \text{avg} = l$
 $\Rightarrow \text{sum} = 5l$

average of all no. $\Rightarrow$

$\boxed{\frac{7d+5l}{12}}$ Ans

[13] If a, b, c, d, e are five consecutive odd numbers, their average is:

(a) $5(a+4)$ (b) $\frac{abcde}{5}$ (c) $5(a+b+c+d+e)$ (d) $a+4$

Soln:- $a, b, c, d, e \rightarrow a, a+2, \boxed{a+4}, a+6, a+8$
 $\uparrow$
 $\text{avg} = \boxed{c} \Rightarrow \boxed{a+4}$

[14] The average of four consecutive even numbers is 15. The 2nd highest number is:

Soln:-

$\text{---} \quad \text{---} \quad 15 \quad 16 \quad 18 \leftarrow \text{highest}$
 $\uparrow$
 avg

2nd Highest = $\boxed{16}$ Ans

15 The average of the first 7 integers in a series of 13 consecutive odd integers is 37. What is the average of the entire series?

Soln:- $\underline{\quad} \quad \underline{\quad} \quad \underline{\quad} \quad \underline{37} \quad \underline{39} \quad \underline{41} \quad \underline{43} \quad \dots \dots \dots 13 \text{ term}$
 $\uparrow$ avg of first 7 integers $\uparrow$ 7th term
 $\text{average} = \frac{13+1}{2} = 7^{\text{th}} \text{ term}$
Avg. = 43 Ans

16 The average of five consecutive odd numbers is 27. What is the product of the first and the last number?

Soln:- $\underline{23} \quad \underline{25} \quad \underline{27} \quad \underline{29} \quad \underline{31}$
 $\uparrow$ avg $\rightarrow 23 \times 31 = \underline{713}$ Ans

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17 The average of seven consecutive positive integers is 26. The smallest of these integers is:

Soln:- $\underline{23} \quad \underline{24} \quad \underline{25} \quad \underline{26} \quad \underline{\quad} \quad \underline{\quad} \quad \underline{\quad}$
 Smallest $\uparrow$ avg

18 The average of eight consecutive odd numbers is 28. The sum of the smallest and the largest number is:

Soln:- $\underline{21} \quad \underline{23} \quad \underline{25} \quad \underline{27} \quad \underline{29} \quad \underline{31} \quad \underline{33} \quad \underline{35}$
 $\uparrow$ 28
 $\text{Sum} = 21 + 35 = \underline{56}$ Ans

- 19] The average of eight successive numbers is 6.5. The average of the smallest and the greatest numbers among them will be:

Soln:- (3), 4, 5, 6, 7, 8, 9, (10)

6.5
(avg)

$$\text{avg} = \frac{3+10}{2} = \boxed{6.5}$$

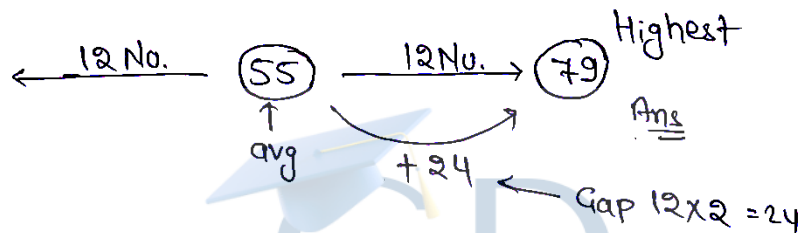
m-n

$$\text{avg} = \frac{\text{first} + \text{Last}}{2}$$

So it will be same $\boxed{6.5}$

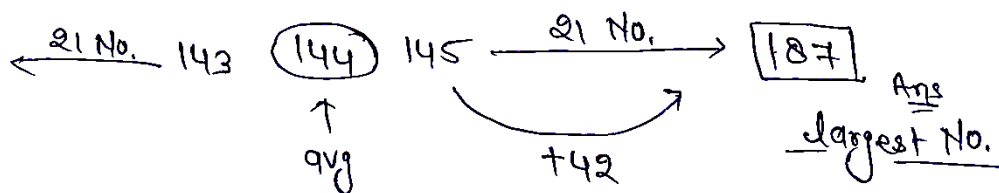
- 20] The average of 25 consecutive odd integers is 55. The highest of these integers is -

Soln:-



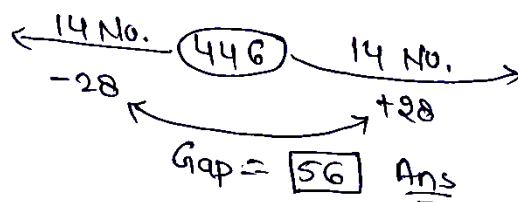
- 21] The average of 44 consecutive odd numbers is 144. What is the largest number?

Soln:-



- 22] If average of 29 consecutive even number is 446, then find difference between first and last number?

Soln



Q3 The average of 5 consecutive natural numbers is M . If the next three natural numbers are also included, how much more than M will the average of these 8 numbers be?

Soln:- 1, 2, (3), 4, 5, 6, 7, 8

$\downarrow$ old avg. $\downarrow$ New avg.

$4.5 - 3 = \boxed{1.5}$ Ans

Q4 The average of 5 consecutive even number is M . If the next five even number are also included, the average of ten number will be:

Soln:- 2, 4, (6), 8, 10, 12, 14, 16, 18, 20

$\uparrow$ old avg. $\uparrow$ New avg.

$\frac{10+12}{2} = 11$

(6) $\xrightarrow{+5}$ (11) New avg = $\boxed{M+5}$ Ans

Q5 The average of 3 consecutive even numbers is A . If next 5 even number are added then what is the average of these 8 numbers?

Soln:- 2, (4), 6, 8, 10, 12, 14, 16

$\uparrow$ old avg. $\xrightarrow{+5}$ New avg.

$\boxed{A+5}$ Ans

⇒

If Average of n numbers
is

$$\boxed{x}$$

In every
term

New Avg.

$$+a$$

⇒

$$x+a$$

$$-a$$

⇒

$$x-a$$

$$\times a$$

⇒

$$ax$$

$$\div a$$

⇒

$$\frac{x}{a}$$

[26] Average of 36 results is 18. If 2 is subtracted from each result, then what will be the new average of the result?

Soln:- New avg = $18 - 2 = \boxed{16}$ Ans

[27] The average of 11 numbers is 7. If every number is doubled, what will be the new average of the numbers?

Soln:- New avg = $7 \times 2 = \boxed{14}$ Ans

[28] The average of 126 numbers is 951. If each number is multiplied by 0.2 and added to 3.6, the average of the new set of numbers is:

Soln:-

$$\text{New avg.} = 951 \times 0.2 + 3.6$$

$$= \boxed{193.8} \text{ Ans}$$

29 The average of three numbers of which greatest is 16 is 12. If the smallest is half of the greatest, the remaining number is:

Soln:-

$$\begin{array}{ccc} a & b & c \\ \text{(smallest)} & & \text{(greatest)} \end{array} \quad 8 + b + 16 = 12 \times 3$$

$$b + 24 = 36$$

$$b = \boxed{12} \text{ Ans}$$

30 The average of first three numbers is double of the fourth number. If the average of all the four number is 12, find the 4th number

Soln:-

$$\begin{array}{cccc} a & b & c & d \\ \text{avg.} \rightarrow 2x & & & x \end{array}$$

$$\text{Total avg} = \frac{2x \times 3 + x}{4} = 12$$

$$\Rightarrow 7x = 48$$

$$x = \boxed{\frac{48}{7}} \text{ Ans}$$

31 The average of three consecutive odd numbers is 52 more than $\frac{1}{3}$ rd of the largest of these number. What is the smallest of these numbers?

Soln:-

$$x, x+2, x+4$$

$$\text{ACQ:} \quad \frac{3x+6}{3} = \frac{(x+4)}{3} + 52$$

$$x+2 = \frac{x+4+156}{3}$$

$$2x = 154 \Rightarrow \boxed{x = 77}$$

$$\text{smallest} = \boxed{77} \text{ Ans}$$

[32] The arithmetic mean of the following numbers:

1, 2, 2, 3, 3, 3, 4, 4, 4, 4, 5, 5, 5, 5, 5, 6, 6, 6, 6, 6, 6, 7, 7, 7, 7, 7, 7 is

Soln:-
$$\frac{1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 + 7^2}{1 + 2 + 3 + 4 + 5 + 6 + 7} = \frac{7 \times 8 \times 15}{6} = \frac{7 \times 8}{2} = \boxed{5} \text{ Ans}$$

[33] If average of 20 observations $x_1, x_2, \dots, x_{20}$ is y , then the average of $x_1 - 101, x_2 - 101, x_3 - 101, \dots, x_{20} - 101$ is

Soln:- $\boxed{y - 101} \text{ Ans}$

[34] The average of n numbers $x_1, x_2, \dots, x_n$ is $\bar{x}$. Then the value of $\sum_{i=1}^n (x_i - \bar{x})$ is equal to

Soln:- New avg = $\bar{x} - \bar{x} = \boxed{0} \text{ Ans}$
 old avg Subtracted (in every term)

[35] If the average of x and $\frac{1}{x}$ ($x \neq 0$) is M , then the average of $x^2 + \frac{1}{x^2}$ is:

Soln

$$\frac{x + \frac{1}{x}}{2} = M$$

$$x + \frac{1}{x} = 2M$$

$$\frac{x^2 + \frac{1}{x^2}}{2} = ?$$

$$\Rightarrow \frac{(2M)^2 - 2}{2}$$

$$= \boxed{2M^2 - 1} \text{ Ans}$$

36 Of the three numbers, the first is twice the second, and the second is twice the third. The average of the reciprocal of the numbers is $\frac{7}{12}$. The numbers are:

Soln:-

I	II	III
$4x$	$2x$	x

$$\frac{\frac{1}{4x} + \frac{1}{2x} + \frac{1}{x}}{3} = \frac{7}{12}$$

$$\frac{7}{4x} = \frac{7}{12} \times 3$$

$$\boxed{x = 1}$$

no. $\rightarrow$ $\boxed{4, 2, 1}$ Ans

37 The average of marks obtained by A and B is 15 less than that of average marks obtained by B and C. If the marks obtained by C is 65, What is the marks obtained by A?

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Soln:-

$$\left(\frac{B+C}{2}\right) - \left(\frac{A+B}{2}\right) = 15$$

$$\frac{C-A}{2} = 15$$

$$[C = 65]$$

$$65 - A = 30$$

$$\boxed{A = 35}$$
 Ans

38 Three numbers are such that if the average of any two of them is added to the third number the sums obtained are 164, 158 and 132 respectively. What is the average of the original three numbers?

Soln:-

$$\frac{a+b}{2} + c = 164$$

$$\frac{b+c}{2} + a = 158$$

$$\frac{c+a}{2} + b = 132$$

$$2 \frac{(a+b+c)}{2} + (a+b+c) = 454$$

$$a+b+c = \boxed{227}$$

$$\text{avg} = \frac{227}{3} = \boxed{75\frac{2}{3}} \text{ Ans}$$

[39] The average of three numbers a, b and c is 2 more than c. The average of a and b is 48. If d is 10 less than c, then the average of c and d is:

Soln:- $\frac{a+b+c}{3} = c+2$, $\frac{a+b}{2} = 48$

$$96+c = 3c+6 \Rightarrow a+b = 96$$

$$\Rightarrow c = 45$$

$$d = 35$$

$$\text{avg of } c \& d = \frac{45+35}{2} = \boxed{40} \text{ Ans}$$

[40] The sum of 17 consecutive numbers is 289. The sum of another 10 consecutive numbers, whose first term is 5 more than the average of the first set of consecutive numbers, is:

Soln:- $\text{Avg} = \frac{289}{17} = 17$

$$\text{Another set} = 22, 23, 24, \dots, 31 \Rightarrow \text{avg} = \frac{22+31}{2} = \boxed{26.5}$$

$$\text{Sum} = 26.5 \times 10 = \boxed{265} \text{ Ans}$$

41 Average age of 7 students of a class is 28 years, Average age of first three students is 30 years. Age of fourth student is 4 years less than the age of fifth student. Ages of last two students is same and is 5 years more than the average age of first three students. What is the average age of fourth and fifth students?

Soln:-

[SSC CHL-2021 Mains]

<u>3 students</u>	4 th	5 th	<u>6th, 7th</u>
avg = 30	$x-4$	x	35, 35

$$90 + x - 4 + x + 70 = 7 \times 28$$

$$2x = 40$$

$$x = 20$$

$$\begin{array}{l} 4^{\text{th}} \Rightarrow 16 \\ 5^{\text{th}} \Rightarrow 20 \end{array} \quad \text{] avg} = \frac{16+20}{2} = \boxed{18} \text{ Ans}$$

42 In a team of 10 persons nine persons spent ₹40 each for their meal and the remaining one spent ₹9 more than the average expenditure of all the 10 persons. The total expenditure for their meal was:

Soln:- let the average expenditure of all = x

$$9 \times 40 + (x+9) = 10x$$

$$9 \times 40 + 9 = 9x$$

$$x = 41$$

$$\text{Total exp.} = 41 \times 10 = \boxed{410} \text{ Ans}$$

[43] Eight members of a club donate Rs 100 each towards a Relief Fund and the President of the club donates Rs. 50 more than the average donation of all (including President) of them. Then the contribution of the president is:

Soln:- let avg of all = x

$$\text{President} = x + 50$$

$$8 \times 100 + (x + 50) = 9x$$

$$800 + 50 = 8x$$

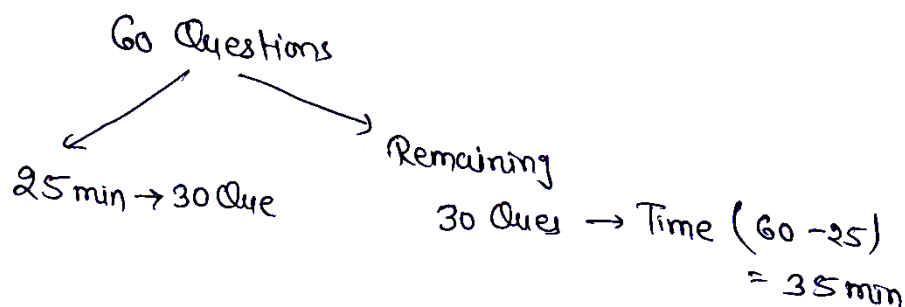
$$x = \frac{850}{8} = 106.25$$

$$\text{President} = x + 50 = 106.25 + 50$$

$$= \boxed{156.25} \text{ Ans}$$

[44] An hour-long test has 60 problems. If a student completes 30 problem in 25 minutes, then the required seconds he has taken on average for computing each of the remaining problem is:

Soln:-



$$\text{Avg} = \frac{35 \text{ min}}{30} \times 60 \leftarrow \text{for sec.}$$

$$= \boxed{70 \text{ second}} \text{ Ans}$$

[45] If the average price of three chairs is Rs 14014 and their prices in the ratio of 3:4:7, then highest price of chair is:

Soln:- Price $\rightarrow 3:4:7$

$$\text{avg} = \frac{3+4+7}{3} = \frac{14}{3} \text{ Unit} \Rightarrow 14014$$

$$1 \text{ Unit} \Rightarrow 3003$$

$$\begin{aligned} \text{highest} &= 7 \text{ Unit} = 7 \times 3003 \\ &= \boxed{21021} \text{ Ans} \end{aligned}$$

[46] Average of n numbers is 'a'. The first number is increased by 2, second one is increased by 4, the third one is increased by 8 and so on. The average of the new numbers is:

Soln:- Total increased $= 2 + 4 + 8 + \dots$

$$= 2^1 + 2^2 + 2^3 + \dots + 2^n$$

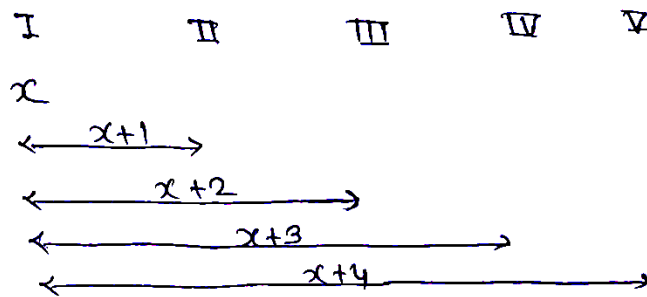
$$= \frac{2(2^n - 1)}{2 - 1}$$

$$\text{increased avg} = \frac{2(2^n - 1)}{n}$$

$$\text{New avg} = \left[a + \frac{2(2^n - 1)}{n} \right] \text{ Ans}$$

47 5 members of a team are weighed consecutively and their average weight calculated after each member is weighed. If the average weight increased by one kg each time, how much heavier is the last player than the first one?

Soln :-



sum of 5 members
Acc $(x+4) \times 5 = 5x + 20$

sum of 4 members
 $(x+3) \times 4 = 4x + 12$

heavier
 $(x+8)$ Ans

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48 A shop of electronic goods remains closed on Monday. The average sales per day for remaining six days of a week is ₹15640 and the average sale of Tuesday to Saturday is ₹14124. The sales on Sunday is:

Soln :-

Total Sales $\rightarrow 15640 \times 6 = 93840$

Total Sales (Tues $\leftrightarrow$ sat.) $= 14124 \times 5 = 70620$

Sunday $= 93840 - 70620$
 $= \boxed{23220}$ Ans

[49] The average temperature of Monday, Tuesday, Wednesday and Thursday is 60° , the average for Tuesday, Wednesday, Thursday and Friday is 63° ; If the ratio of temperature for Monday and Friday is $21:25$, then what is the temperature of Friday?

Soln:-

$$\begin{aligned} Mo + Tu + We + Th + &= 60 \times 4 \\ Tu + We + Th + Fr &= 63 \times 4 \end{aligned}$$

$$- \quad Mo - Fr = -12$$

$$21x - 25x = -12$$

$$\boxed{x = 3}$$

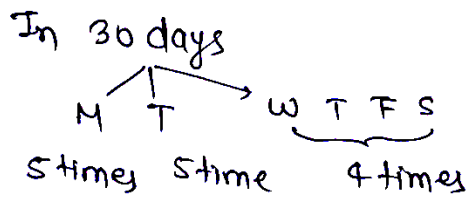
$$\text{Friday} = 25x = 25 \times 3 = \boxed{75^\circ} \text{ Ans}$$

[50] A library has an average of 265 visitors on Sundays and 130 visitors on other days. The average number of visitors per day in a month of 30 days beginning with a Monday is:

Soln:- Total Sunday = 4

$$\text{Total visitors on Sunday} = 4 \times 265 = 1060$$

In 30 days



$$\begin{aligned} \text{Total visitors} &= (5 + 5 + 4 \times 4) \times 130 \\ &= 26 \times 130 = 3380 \end{aligned}$$

$$\text{avg of per day} = \frac{3380 + 1060}{30} = \frac{4440}{30} = \boxed{148} \text{ Ans}$$

Concept

No. of data $\rightarrow n_1 \quad n_2 \quad n_3 \quad n_4 \dots$

Average $\rightarrow x_1 \quad x_2 \quad x_3 \quad x_4 \dots$

$$\left[\text{Net Avg / weighted Avg} \Rightarrow \frac{n_1 x_1 + n_2 x_2 + n_3 x_3 + \dots}{n_1 + n_2 + n_3 + \dots} \right]$$

[51] The frequency distribution data is given below. If the average age is 17 years, the value of m is :

Age (in years) : 8 20 26 29

No. of People : 3 2 m 1

Soln:-

$$\text{avg} = \frac{8 \times 3 + 20 \times 2 + 26 \times m + 29 \times 1}{3 + 2 + m + 1}$$

$$17 = \frac{24 + 40 + 26m + 29}{6 + m}$$

$$102 + 17m = 93 + 26m$$

$$9m = 9$$

$$\boxed{m = 1} \text{ Ans}$$

[52] If out of 10 selected students for an examination, 3 were of 20 years, age, 4 of 21 and 3 of 22 years, the average age of the group is:

Soln:-

$$\begin{aligned} \text{avg} &= \frac{3 \times 20 + 4 \times 21 + 3 \times 22}{10} \\ &= \frac{60 + 84 + 66}{10} \Rightarrow \frac{210}{10} = \boxed{21 \text{ years}} \quad \underline{\underline{\text{Ans}}} \end{aligned}$$

[53] Numbers of boys and girls are 'x' and 'y' respectively. Ages of a girl and a boy are 'a' years and 'b' years respectively. The average age of all boys and girls.

Soln:-

$$\text{avg} = \frac{ax + by}{x + y} \quad \underline{\underline{\text{Ans}}}$$

[54] The ratio of the number of players in the three cricket team A, B and C is 2:5:3. If the ratio of number of runs scored per player for each of the three teams A, B and C is 30:17:25 respectively, then what is the average number of runs scored per player across all the three team collectively?

Soln:-

$$\begin{aligned} \text{Avg} &= \frac{30 \times 2 + 17 \times 5 + 25 \times 3}{2 + 5 + 3} \\ &= \frac{60 + 85 + 75}{10} \\ &= \frac{220}{10} = \boxed{22} \quad \underline{\underline{\text{Ans}}} \end{aligned}$$

55 The average weight of A, B and C is 65 kg. If the average weight of C and B is 61.5 kg, and the average weight of A and C is 68.5 kg. then the weight of C is:

Soln:-

$$A+B+C = 65 \times 3 = 195 \text{ kg}$$

$$B+C = 61.5 \times 2 = 123 \text{ kg}$$

$$A+C = 68.5 \times 2 = 137$$

$$C = (123 + 137) - 195 \Rightarrow \boxed{65} \text{ Ans}$$

56 The average weight of Subha, Govinda and Reshma is 65 kg. If the average weight of Subha and Govinda be 63 kg and that of Govinda and Reshma be 70 kg. then the weight of Govinda is -

Soln:-

$$\text{Govinda} \Rightarrow (70 \times 2 + 63 \times 2) - (65 \times 3)$$

$$\Rightarrow 140 + 126 - 195 \Rightarrow \boxed{71 \text{ kg}} \text{ Ans}$$

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57 The average of A, B and C is 18 and that of C, D and E is 12 and that of E and F is 6.5 and that of E and C is 3.5. What is the average of A, B, C, D, E, and F?

Soln:-

$$A+B+C = 18 \times 3 = 54$$

$$C+D+E = 12 \times 3 = 36$$

$$E+F = 6.5 \times 2 = 13$$

$$E+C = 3.5 \times 2 = 7$$

$$+ \quad \underline{A+B+C+D+E+F+2(E+C)} = 110$$

$$A+B+C+D+E+F = 110 - 2 \times 7 = 96$$

$$\text{avg} = \frac{96}{6} = \boxed{16} \text{ Ans}$$

[58] The average temperature for Monday, Wednesday and Friday was 41°C . The average temperature for Wednesday, Thursday and Friday was 42°C . If the temperature on Thursday was 43°C , then the temperature on Monday was:

Soln:-

$$\begin{aligned} (M + W + F) &= 41^{\circ}\text{C} \times 3 \\ (W + Th + F) &= 42^{\circ}\text{C} \times 3 \\ \hline - \quad Th - M &= 1 \times 3 \quad [\because Th = 43^{\circ}\text{C}] \\ 43^{\circ}\text{C} - 3 &= \text{Monday} \\ \text{Monday} &= \boxed{40^{\circ}\text{C}} \quad \underline{\text{Ans}} \end{aligned}$$

[59] The average salary of a person for the months of February, March, April and May is Rs. 12,000 and that for the months March, April, May, Jun is Rs 3,000. If his salary for the month of June is Rs. 9,000, then what is his salary for the month of February?

Soln:-

$$\begin{aligned} \text{Feb} + \text{March} + \text{April} + \text{May} &= 12000 \times 4 \\ \text{March} + \text{April} + \text{May} + \text{Jun} &= 3000 \times 4 \\ \hline - \quad \text{Feb} - \text{Jun} &= (12000 - 3000) \times 4 \\ \text{Feb} - 9000 &= 9000 \times 4 \\ \text{Feb} &= \boxed{45,000} \quad \underline{\text{Ans}} \end{aligned}$$

[60] The average of 40 numbers is 36. The average of the first 25 number is 31 and the average of last 16 numbers is 43. Find the 25th number?

Soln m-I Basic:-

$$\text{Sum of 40 No.} \Rightarrow 36 \times 40 = 1440$$

$$\text{Sum of first 25 No.} \Rightarrow 31 \times 25 = 775$$

$$\text{Sum of last 16 No.} \Rightarrow 43 \times 16 = 688$$

$$25^{\text{th}} \text{ No.} \Rightarrow (775 + 688) - 1440 = \boxed{23} \text{ Ans}$$

m-II Deviation:-

$$-5 \times 25 + 7 \times 16$$

$$\Rightarrow -125 + 112 = \boxed{-13}$$

$$25^{\text{th}} \text{ No.} = 36 - 13 = \boxed{23} \text{ Ans}$$

[61] The average revenues of 7 consecutive years is Rs. 83 lakhs. The average of first 4 years is 78 lakhs and that of last 4 years is Rs. 90 lakhs, find the revenue for the 4th year.

Soln:- $4^{\text{th}} \text{ year} \Rightarrow (4 \times 78 + 4 \times 90) - (7 \times 83)$

$$\Rightarrow 672 - 581 \Rightarrow \boxed{91 \text{ lakhs}} \text{ Ans}$$

m-II Deviation:-

$$-5 \times 4 + 7 \times 4$$

$$\Rightarrow -20 + 28 = \boxed{+8}$$

$$4^{\text{th}} \text{ year} = 83 + 8$$

$$= \boxed{91 \text{ lakhs}} \text{ Ans}$$

62 Out of four numbers the average of the first three is 16 and that of the last three is 15. If the last number is 20 then the first number is:

Soln:-

$$a+b+c = 16 \times 3$$

$$b+c+d = 15 \times 3$$

$$a-d = 3$$

$$a = 3+20 = \boxed{23} \text{ Ans}$$

63 In a set of three numbers, the average of the first two numbers is 7, the average of the last two numbers is 10, and the average of the first and the last number is 14. What is the average of the three numbers?

Soln:-

$$a+b = 7 \times 2$$

$$b+c = 10 \times 2$$

$$a+c = 14 \times 2$$

$$\& (a+b+c) = (7+10+14) \times 2$$

$$a+b+c = 31 \quad \text{avg} = \boxed{\frac{31}{3}} \text{ Ans}$$

64 The average of 25 numbers is 64. The average of the first 13 numbers and that of the last 13 numbers are 62.8 and 72.2 respectively. If the 12th number is 61 and if the 12th and 13th numbers are excluded, then what is the average of the remaining number (correct to one decimal place)?

Soln:-

Deviation:- $-1.2 \times 13 + 8.2 \times 13 = +91$

$$13^{\text{th}} \text{ No.} = 64 + 91 = 155$$

$$12^{\text{th}} \text{ No.} = 61$$

$$\begin{aligned} \text{after excluding Sum of 23 No.} &= 25 \times 64 - 155 - 61 \\ &= 1600 - 155 - 61 = 1384 \end{aligned}$$

$$\text{avg} = \frac{1384}{23} \approx \boxed{60.2} \quad \underline{\underline{\text{Ans}}}$$

[65] The average of 33 numbers is 74. the average of the first 17 numbers is 72.8 and that of the last 17 numbers is 77.2. If the 17th number is excluded, then what will be the average of the remaining numbers (Correct to one decimal place) ?

Soln :- $17^{\text{th}} \text{ No.} \Rightarrow -1.2 \times 17 + 3.2 \times 17$

Deviation $\Rightarrow \boxed{+34}$

$17^{\text{th}} \text{ No.} \Rightarrow 74 + 34 \Rightarrow 108$

avg. of 32 No. = $\frac{33 \times 74 - 108}{32} = \boxed{72.9} \quad \underline{\underline{\text{Ans}}}$

[66] The average of 12 numbers is 39. The average of the last five numbers is 35 and that of first four numbers is 40. The fifth number is 6 less than the sixth number and 5 more than the seventh number. The average of the sixth and seventh numbers is ;

Soln :- Deviation method :-

$$\begin{array}{ccccc} & 5^{\text{th}} & 6^{\text{th}} & 7^{\text{th}} & \\ & \uparrow & \uparrow & \uparrow & \\ (-4) \times 5 & + 4 \times 1 & + x - 6 & + x & + x - 11 = 3 \times 39 \end{array}$$

$$-20 + 4 + x - 6 + x + x - 11 = 117$$

$$3x = 150$$

$$x = 50$$

$6^{\text{th}} \text{ No.} = 50$

$7^{\text{th}} \text{ No.} = 39 \quad \text{avg} = \frac{50 + 39}{2} = \boxed{44.5} \quad \underline{\underline{\text{Ans}}}$

[67] The average of sixteen numbers is 48. The average of the first six of these numbers is 45 and that of the last seven numbers is 53. The seventh and the eighth numbers are respectively 3 and 7 greater than the ninth number. what is the average of the ninth and seventh numbers?

Soln:-

$$-3 \times 6 + 5 \times 7 + \overset{8^{th}}{(x+7)} + \overset{7^{th}}{(x+3)} + \overset{9^{th}}{x} = 3 \times 48$$

$$-18 + 35 + 3x + 10 = 144$$

$$3x = 117$$

$$x = 39$$

$$7^{th} = 39 + 3 = 42$$

$$9^{th} = 39$$

$$avg = \frac{42 + 39}{2} = \boxed{40.5} \underline{\underline{Ans}}$$

[68] The average of thirteen numbers is 47. The average of the first three numbers is 39, and that of next seven numbers is 49. The 11th number is two times the 12th number and 12th number is 3 less than the 13th number. what is the average of 11th and 13th number?

Soln:-

$$-8 \times 3 + 2 \times 7 + \overset{11^{th}}{2x} + \overset{12^{th}}{x} + \overset{13^{th}}{x+3} = 3 \times 47$$

$$-24 + 14 + 4x + 3 = 141$$

$$4x = 148$$

$$x = 37$$

$$11^{th} = 74$$

$$13^{th} = 40$$

$$avg = \frac{74 + 40}{2}$$

$$= \frac{114}{2} = \boxed{57} \underline{\underline{Ans}}$$

69 The average of twelve numbers is 55.5. The average of the first four numbers is 53.4 and that of the next four numbers is 54.6. The 10th number is greater than the 9th number by 3 but lesser than the 11th and 12th numbers by 2 and 3 respectively. What is the average of the 10th and the 12th numbers?

Soln:-

$$-2.1 \times 4 - 0.9 \times 4 + \overset{9^{th}}{x} + \overset{10^{th}}{(x+3)} + \overset{11^{th}}{(x+5)} + \overset{12^{th}}{(x+6)} = 4 \times 55.5$$

$$-8.4 - 3.6 + 4x + 3 + 5 + 6 = 222$$

$$4x = 222 - 2$$

$$x = 55$$

$$10^{th} = 58$$

$$12^{th} = 61$$

$$\text{avg} = \frac{58+61}{2} = \boxed{59.5} \text{ Ans}$$

70 Average age of 7 students of a class is 28 years. Average age of first three students is 30 years. Age of fourth student is 4 years less than the age of fifth student. Age of last two students is same and is 5 years more than the average age of first three students. What is the average age of fourth and fifth student?

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Soln:-

$$3 \times 30 + \overset{4^{th}}{(x-4)} + \overset{5^{th}}{x} + \overset{6^{th}}{35} + \overset{7^{th}}{35} = 7 \times 28$$

$$2x = 40$$

$$x = 20$$

$$4^{th} = 16, 5^{th} = 20$$

$$\text{avg} = \frac{16+20}{2}$$

$$= \boxed{18 \text{ years}}$$

71 In the month of March Dalip earned on an average Rs. 501 per day. For the first 18 days, his average earning was Rs. 495 and for the last 15 days his average earning was Rs. 505. On 18th March he earned Rs. 50 more than that earned on 17th March. How much did he earn on 18th March?

Soln:- Basic:-

$$\text{Earning of 17th + 18th day} = (495 \times 18 + 505 \times 15) - (501 \times 31)$$

$$x + x + 50 = 8910 + 7575 - 15531$$

$$2x + 50 = 954$$

$$x = 452$$

$$18^{\text{th}} \text{ day earning} = 452 + 50 = \boxed{502} \text{ Ans}$$

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Qn-II Deviation:-

$$-6 \times 18 + 4 \times 15 \Rightarrow -108 + 60 \Rightarrow \boxed{-48}$$

$$\begin{aligned} 17^{\text{th}} + 18^{\text{th}} \text{ No.} &= 2 \times 501 - 48 \\ &= 1002 - 48 \end{aligned}$$

$$x + x + 50 = 954$$

$$2x = 904$$

$$x = 452$$

$$18^{\text{th}} \Rightarrow \boxed{502} \text{ Ans}$$

7Q Ras Bihari a plumber, earned on an average Rs 925 per day in the month of January. He earned on an average Rs. 881 per day during the first 20 days and Rs. 915 per day during the last 20 days. What was his average income per day from 12th January to 20th January?

Soln:-

Deviation:-

$$-44 \times 20 - 10 \times 20$$

$$\Rightarrow -880 - 200 = \boxed{-1080}$$

$$\text{Sum of 12th to 20th Jan.} \Rightarrow 9 \times 925 - 1080$$

$$\Rightarrow 8325 - 1080$$

$$= 7245$$

$$\text{avg} = \frac{7245}{9} = \boxed{805} \text{ Ans}$$

m-II Basic:

Total

$$\text{Sum in January} = 925 \times 31 = 28675$$

$$\text{Earning of first 20 days} = 881 \times 20 = 17620$$

$$\text{Earning of Last 20 days} = 915 \times 20 = 18300$$

$$\text{Earning of 12th to 20th Jan} = (17620 + 18300) - 28675$$

$$= 35920 - 28675$$

$$= 7245$$

$$\text{avg} = \frac{7245}{9}$$

$$= \boxed{805} \text{ Ans}$$

73 The average age of a family with 5 members is 28. If one of the member of age 20 is excluded the average age of the family becomes.

Soln:- Basic:-

$$\text{Total age} = 5 \times 28 = 140$$

$$\begin{aligned} \text{Remaining Age} &= 140 - 20 \\ &= 120 \end{aligned}$$

$$\text{avg} = \frac{120}{4} = \boxed{30 \text{ year}} \quad \underline{\text{Ans}}$$

m-II Deviation method:-

$$28 - 20 = 8$$

$$\text{effect} = \frac{8}{4} = +2$$

$$\begin{aligned} \text{New Avg} &= 28 + 2 \\ &= \boxed{30 \text{ years}} \end{aligned}$$

74 The average monthly salary of 19 members of a group is ₹ 16,000. If one more member whose monthly salary is ₹ 20,000 has joined the group then the average salary of the group is - **MathswithGajanand**

Soln:- Basic

$$\begin{aligned} \text{Total salary of 19 members} \\ &= 19 \times 16000 \\ &= 3,04,000 \end{aligned}$$

$$\begin{array}{r} \text{1 Member} = 20,000 \\ \text{Add} \quad + \\ \hline \text{total} = 3,24,000 \end{array}$$

$$\text{avg} = \frac{3,24,000}{20}$$

$$= \boxed{16,200} \quad \underline{\text{Ans}}$$

m-II

$$\begin{aligned} \text{Extra salary} &= 20000 - 16000 \\ &= 4000 \end{aligned}$$

$$\begin{aligned} \text{which will be divided among} \\ \text{20 members} &= \frac{4000}{20} = 200 \end{aligned}$$

$$\begin{aligned} \text{New Avg} &= 16000 + 200 \\ &= \boxed{16,200} \quad \underline{\text{Ans}} \end{aligned}$$

75 Given that the mean of five numbers is 28. If one of them is excluded, the mean gets reduced by 5. Determine the excluded number.

Soln:- **m-I** Basic

$$\text{Sum of all 5 number} = 28 \times 5 \\ = 140$$

$$\text{Sum of all 4 No.} = 23 \times 4 = 92$$

$$\text{excluded no.} = 140 - 92 \\ = \boxed{48} \text{ Ans}$$

(Remaining 4 Numbers)

m-II $5 \times \overset{\downarrow}{4} = 20$

$$\text{excluded no.} = 28 + 20 \\ = \boxed{48}$$

76 The average age of 24 boys and their teacher is 15 yr. when the teacher's age is excluded, the average age decreases by 1 year. The age of teacher is:

Soln:- **m-I** Basic

$$\text{Total Age students + Teacher} \\ = 25 \times 15 = 375$$

$$\text{Remaining Age of 24 boys} \\ = 24 \times 14 = 336$$

$$\text{Teacher} = 375 - 336 \\ = \boxed{39 \text{ year}}$$

m-II $\begin{matrix} \text{No of Boys} & \text{decrease} \\ \uparrow & \downarrow \\ \text{effect} = 24 \times 1 = 24 \end{matrix}$

$$\text{Teacher} = 15 + 24 \\ = \boxed{39 \text{ year}} \text{ Ans}$$

77 The average age of eleven cricket players is 20 years. If the age of the coach is also included, the average age increased by 2. The age of the coach is:

Soln:- Basic

$$\begin{aligned}\text{Age of all cricketers} &= 11 \times 20 \\ &= 220\end{aligned}$$

$$\begin{aligned}\text{Age of crickets + coach} &= 12 \times 22 \\ &= 264\end{aligned}$$

$$\begin{aligned}\text{Coach} &= 264 - 220 \\ &= \boxed{44 \text{ years}} \text{ Any}\end{aligned}$$

(11 Cricketer + Coach)
↓
m-II
 $+2 \times 12 = +24$
 $\text{Coach} = 20 + 24$
 $= \boxed{44 \text{ years}}$

78 The average weight of 12 crewmen in a boat is increased by $\frac{1}{3}$ kg, when one of the crewmen whose weight is 55 kg. is replaced by a new man. what is the weight of that new man?

Soln:-

$$\text{Total increase in weight} = 12 \times \frac{1}{3} = 4 \text{ kg}$$

$$\begin{aligned}\text{weight of new man} &= 55 + 4 \text{ kg} \\ &= \boxed{59 \text{ kg}} \text{ Any}\end{aligned}$$

79 Out of 10 teachers of a school one teacher retires and at his place a new teacher of age 25 years joins. As a result of it, the average age of the teachers is reduced by 3 years. The age of the retired teachers is?

Soln:- Age of Retired teacher = $25 \text{ years} + 10 \times 3$
 $= 25 + 30 = \boxed{55 \text{ years}}$ Ans

80 From a class of 24 boys, a boy, aged 10 years, leaves the class and in his place a new boy is admitted. As a result the average age of the class is increased by 2 months. what is the age of the new boy?

Soln:- New boy age = $10 + 24 \times 2 \text{ month}$
 $= 10 + 4 \text{ years} = \boxed{14 \text{ years}}$ Ans

81 A group of boys has an average weight of 36 kg. One boy weight 42 kg leaves the group and another boy weight 30 kg join the group. If the average now become 35.7 kg, then how many boys are there in the group?

Soln:- change in weight = $42 - 30 = 12 \text{ kg}$

chang in Avg. = $36 - 35.7 = 0.3$

No. of Boys = $\frac{12}{0.3} = \boxed{40}$ Ans

82 The average weight of 40 children of a class is 36.2 kg. When three more children with weight 42.3 kg, 39.7 kg and 39.5 kg join the class, the average weight of the 43 children in the class is -

Soln :- Basic :-

$$\begin{aligned} \text{Total weight of 40 children} \\ = 40 \times 36.2 = 1448 \end{aligned}$$

$$\begin{aligned} \text{Total weight of 43 children} \\ = 1448 + 42.3 + 39.7 + 39.5 \\ = 1569.5 \text{ kg} \end{aligned}$$

$$\text{avg} = \frac{1569.5}{43} = \boxed{36.5 \text{ kg}} \text{ Ans}$$

m-II Deviation :-

$$36.2 \text{ kg} \rightarrow 40 \text{ children}$$

$$42.2, 39.7, 39.5 \\ (+6.2), (+3.5), (+3.3)$$

$$\text{Total extra} = 13$$

$$\begin{aligned} \text{New avg} &= 36.2 + \frac{13}{43} \\ &= \boxed{36.50 \text{ kg}} \text{ Ans} \end{aligned}$$

83 The average weight of a certain number of students in a class is 68.5 kg. If 4 new students having weights 72.2 kg, 70.8 kg, 70.3 kg and 66.7 kg join the class, then the average weight of all the students increases by 300 gm. The number of students in the class, initially is :-

Soln :- change/Deviation in weight $\Rightarrow 3.7 + 1.8 + 2.3 + (-1.8)$

$$= 6$$

$$\text{average change} = 300 \text{ gm} = 0.3 \text{ kg}$$

$$\text{No. of students} = \frac{6}{0.3} = \boxed{20}$$

$$\text{initially} = 20 - 4 = \boxed{16} \text{ Ans}$$

84 The mean of 20 items is 55. If two items 45 and 30 are removed, the new mean of the remaining item is:

Soln:- Total sum = $20 \times 55 = 1100$
Remaining sum = $1100 - (45 + 30)$
 $= 1025$
Avg. = $\frac{1025}{18} = \boxed{56.9}$ Ans

m-11

Deviation $\rightarrow -10 - 25 = -35$

Effect = $\frac{35}{18}$

New Avg = $55 + \frac{35}{18}$
 $= \boxed{56.9}$ Ans

85 Average age of 8 men is increased by 3 years when two of them whose age are 30 and 34 years are replaced by 2 person. What is the average age of the 2 person?

Soln:- Total increase in Age = $8 \times 3 = 24$ years
Sum of 2 person (Age) = $30 + 34 + 24 = 88$ years
Avg = $\frac{88}{2} = \boxed{44 \text{ years}}$

86 The average age of 11 players of a cricket team is increased by 2 months when two of them aged 18 years and 20 years are replaced by two new players. The average age of the new players is:

Soln:- Total increased = $11 \times 2 \text{ month} = 22 \text{ month} = 1 \text{ yr } 10 \text{ month}$
Sum of 2 players = $18 + 20 + 1 \text{ yr } 10 \text{ month}$
 $= 39 \text{ Years } 10 \text{ month}$
avg = $\frac{39 \text{ yr } 10 \text{ m}}{2} = \boxed{19 \text{ Year } 11 \text{ month}}$ Ans

87 The average age of A and B is 20 years. If A is to be replaced by C, the average would be 19 years. The average age of C and A is 21 years. The ages of A, B and C in order are:

Soln:-

$$\begin{array}{rcl} A+B & = & 40 \text{ years} \\ - C+B & = & 38 \text{ year} \\ \hline A-C & = & 2 \text{ years} \\ A+C & = & 42 \text{ years} \end{array} > A=22, C=20 \text{ \& B=18 } \underline{\text{Ans}}$$

88 A student finds the average of 10, 2-digit numbers. If the digits of one of the numbers is interchanged, the average increases by 3.6. The difference between the digits of the 2-digit number is:

Soln:- Total increased = $3.6 \times 10 = 36$

then $(10y+x) - (10x+y) = 36$

$$9(y-x) = 36$$

$$y-x = \boxed{4} \text{ Ans}$$

89 The average age of 25 men is 28 years. 5 new men of an average age of 25 years joined them. Find the average age of all the men together.

Soln:-

$$\begin{aligned} \text{New avg} &= 28 - \frac{3 \times 5}{(25+5)} \\ &= 28 - 0.5 = \boxed{27.5} \text{ Ans} \end{aligned}$$

90 The average marks obtained by 40 students of a class is 86. If the 5 highest marks are removed, the average reduces by 1 mark. The average marks of the top 5 students is:

Soln:- Total marks of 40 students
 $= 40 \times 86 = 3440$
Total marks of 35 students
 $= 35 \times 85 = 2975$
Total marks of 5 students
 $= 3440 - 2975 = 465$
 $\text{avg} = \frac{465}{5} = \boxed{93} \text{ Ans}$

m-II

Total marks of 5 students
 $= 86 \times 5 + 35 \times 1$
 $= 465$
 $\text{avg} = \frac{465}{5} = \boxed{93} \text{ Ans}$

91 A team of 8 persons joins in a shooting competition. The best marksman scored 85 points. If he had scored 92 points, then the average score for the team would have been 84. The number of points the team scored was.

Soln:- Let the average of team = x

$$\text{Total score} = 8x$$

$$8x - 85 + 92 = 8 \times 84$$

$$8x = 672 - 7$$

$$8x = \boxed{665} \text{ Ans}$$

92 The average weight of a group of 3 people A, B and C is 70 kg. When D joins this group, the average becomes 60 kg. A man E, whose weight is 5 kg more than of D, replaces A and the average weight of B, C, D and E now becomes 59 kg. What is the average weight of A, D and E? (Correct to the nearest integer)

Soln:- $A+B+C = 70 \times 3 = 210$ — ①
 $A+B+C+D = 60 \times 4 = 240$ — ②

$\boxed{D = 30}$
 $\boxed{E = 35 \text{ kg}}$

$$B+C+D+E = 236$$

$$B+C+30+35 = 236$$

$$B+C = 171 \text{ — ③}$$

from ① & ③ $A = 210 - 171 = \boxed{39 \text{ kg}}$

$$\text{avg} = \frac{A+D+E}{3} \Rightarrow \frac{39+30+35}{3} = \frac{104}{3} \Rightarrow 34.6 \approx \boxed{35 \text{ kg}} \text{ Ans}$$

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93 The average age of 40 students of a class is 16 years. After admission of 10 new students to the class, the average becomes 15 years. If the average age of 5 of the new students is 11 years, then the average age of the remaining 5 new students is:

Soln:- $\Sigma 40 \text{ students} = 40 \times 16 = 640$

$$\Sigma 50 \text{ students} = 50 \times 15 = 750$$

$$\text{Sum of all 10 students} = 750 - 640 = 110$$

$$\text{Sum of 5 new students} = 5 \times 11 = 55$$

$$\text{So Required answer} = \frac{110-55}{5} = \frac{55}{5} = \boxed{11} \text{ Ans}$$

[94] The average of 35 consecutive natural numbers is N .

Dropping the first 10 numbers and including the next 10 numbers, the average is changed to M . If the value of $M^2 - N^2 = 600$, then the average of $3M$ and $5N$ is:

Soln:- Let the numbers

$$x+1, x+2, \dots, x+35$$

$$\text{So } (x+1) + (x+2) + \dots + (x+35) = 35N$$

$$35x + \frac{35 \times 36}{2} = 35N$$

$$x + 18 = N$$

Now

$$x+11, x+12, \dots, x+45 = M \times 35$$

$$35x + (11+12+\dots+45) = M \times 35$$

$$35x + \frac{35}{2}(11+45) = 35M$$

$$x + 28 = M$$

Given

$$(x+28)^2 - (x+18)^2 = 600$$

$$(2x+46)(10) = 600$$

$$x = 7$$

$$N = x+18 = 25$$

$$M = x+28 = 35$$

$$\text{Avg} = \frac{3 \times 35 + 5 \times 25}{2}$$

$$= \frac{230}{2} = \boxed{115} \text{ Ans}$$

[95] Average age of 20 student is 21 years. 2 students leave the group and 1 new student joins the group. The average now becomes 20 years. If age of one of the student who left the group is 26 years, then what is the age of other student who left the group?

Soln:- Total age of 20 students = $21 \times 20 = 420$
 Total age of 19 students = $19 \times 20 = 380$

$$420 - x - y + N = 380$$

$$x + y - N = 40$$

$$26 + y - 20 = 40$$

$$y = \boxed{34 \text{ years}} \text{ Ans}$$

[96] If 40 added to a list of natural numbers, the average is increased by 4. When 30 is added to the new list, the average of the numbers in the new list is increased by 1. How many numbers were in the original list?

Soln:- Let be the number = x and avg = a

$$ax + 40 = (a + 4)(x + 1)$$

$$ax + 40 = ax + a + 4x + 4$$

$$a + 4x = 36 \text{ --- (1)}$$

$$(ax + 40 + 30) = (a + 5)(x + 2)$$

$$70 = 2a + 5x + 10$$

$$2a + 5x = 60 \text{ --- (2)}$$

from (1) & (2)

$$\boxed{x = 4} \text{ Ans}$$

The number will be $\boxed{4}$

97 Three year ago, the average age of a husband, wife and child was 26 year, and that of the wife and the child 5 years ago was 20 years. The present age of the husband is:

Solⁿ:- Present age of $H + W + C = (26 + 3) \times 3 = 87$ years
 $W + C = (20 + 5) \times 2 = 50$ years
Husband = $87 - 50 = \boxed{37 \text{ years}}$ Ans

98 The average age of a husband and wife, who were married 4 years ago was 25 years at the time of their marriage. The average age of the family consisting of husband, wife and a child, born during the interval is 20 years today. The age of the child is:

Solⁿ:- Present Age of
Husband + wife = $(25 + 4) \times 2 = 58$ years
Husband + wife + son = $20 \times 3 = 60$ years
Son = $60 - 58 = \boxed{2 \text{ years}}$ Ans

99 The average age of 6 members of a family is 25 years. If the youngest member of the family is 15 year old, then what was the average age of the family at the time of the birth of the youngest member?

Solⁿ:- Total sum of 6 member's age = $6 \times 25 = 150$
Sum of 5 member's age = $150 - 15 = 135$
avg = $\frac{135}{5} = 27$ years
15 year ago avg = $27 - 15 = \boxed{12 \text{ years}}$ Ans

100 3 year ago, the average age of a family of 5 members was 17 years. A baby having been born, the average age of the family is same today. The present age of the baby is:

Soln:- Avg age of family in present = $17 + 3 = 20$ year

$$\text{Sum} = 20 \times 5 = 100 \text{ year}$$

$$\text{after baby, sum of family} = 17 \times 6 = 102 \text{ year}$$

$$\text{Baby} = 102 - 100 = \boxed{2 \text{ year}} \text{ Ans}$$

101 There is a group of 8 teachers. One teacher leaves the group and a new teacher joins the group. Due to this the average age of teachers becomes same as the average 2 year ago. If the member who left was aged 42 then what is the age of new teacher?

Soln:- New teacher's age = $42 - \overset{\substack{\uparrow \\ \text{No. of teacher}}}{8} \times 2$ less 2 year

$$= \boxed{26 \text{ year}} \text{ Ans}$$

102 B was born when A was 4 year 7 month old and C was born when B was 3 year 4 month old. When C was 5 year 2 month old, then their average age was:

Soln:-

$$\begin{array}{r} A - B = 4y 7m \\ + B - C = 3y 4m \\ \hline A - C = 7y 11m \end{array}$$

when $C = 5y 2m$, $A = 13y 1m$
 $B = 8y 6m$

$$\text{avg} = \frac{A+B+C}{3} = \frac{26y 9m}{3} = \boxed{8y 11 \text{ month}} \text{ Ans}$$

103 A batsman in his 11th inning makes a score of 77 runs thereby increasing in his average scores by 3. What is his average score after the 11th inning?

Solⁿ:- Basic:-

Let the average of 10 inning = x

$$10x + 77 = (x+3) \times 11$$

$$77 = x + 33$$

$$x = 44$$

avg after 11th inning = $(x+3)$

$$= \boxed{47} \text{ Ans}$$

Short Trick

$$77 - 3 \times 10$$

$$77 - 30 = \boxed{47} \text{ Ans}$$

104 The average of runs scored by a cricketer in his 99 innings is 99. How many runs will he have to score in his 100th innings so that his average of runs in 100 innings may be 100?

Solⁿ:- Basic Run in 100th inning = x

$$\frac{99 \times 99 + x}{100} = 100$$

$$9801 + x = 10,000$$

$$x = \boxed{199} \text{ Ans}$$

Short Trick

$$\frac{99, 99, 99, \dots}{99 \text{ Inning}} \quad \frac{\quad}{100^{\text{th}} \text{ Inning}}$$

↓

$$99 + 100$$

$$\boxed{199} \text{ Ans}$$

105 The average of runs of a cricket player of 20 innings was 32. How many runs must he make in his next innings so as to increase his average of runs by 4?

Soln:- Basic

$$20 \times 32 + x = 21 \times (32 + 4)$$

$$640 + x = 756$$

$$x = \boxed{116} \text{ Ans}$$

Short Trick

$$\text{Next inning} = 32 + 21 \times 4$$

$$= 32 + 84$$

$$= \boxed{116} \text{ Ans}$$

106 A scored 73, 76, 20 and 7 runs in four out of five innings. What should be his score in fifth innings. If he has to make an average of 55 run in five innings.

Soln:-

$$\frac{73 + 76 + 20 + 7 + x}{5} = 55$$

$$176 + x = 275$$

$$x = \boxed{99} \text{ Ans.}$$

107 In a 20 over match, the required run rate to win is 7.2. If the run rate is 6 at the end of the 15th over, the required run rate to win the match is:

Soln:-

$$\text{Total run} = 20 \times 7.2 = 144$$

$$\text{run in 15th over} = 15 \times 6 = 90$$

$$\text{Required run} = 144 - 90 = 54 \text{ runs}$$

$$\text{Run rate} = \frac{54}{5} = \boxed{10.8} \text{ Ans}$$

108 In a one day match of 50 overs in an innings the team A had a run rate of 5.3 runs per over. Team B is playing and 5 overs are left and the required run rate to tie the match is 7.2 per over to match the score of team A. what is team B's score?

Soln:-

$$\text{Run scored by Team A} = 50 \times 5.3 = 265$$

$$\text{Run scored by Team B in 5 overs} = 7.2 \times 5 = 36$$

$$\text{Remaining Run} = 265 - 36 = \boxed{229} \text{ Ans}$$

109 A cricketer whose bowling average is 12.4 runs per wicket takes 5 wickets for 26 runs in the next innings and thereby decreases his average by 0.4. The number of wickets taken by him till the last match was:

Soln:- Alligation:-

$$\begin{array}{rcc}
 12.4 & & 5.2 \quad \leftarrow \frac{26}{5} \\
 & \swarrow & \searrow \\
 & 12 & \\
 & \swarrow & \searrow \\
 12.4 - 0.4 \rightarrow & 6.8 & : \quad 0.4 \\
 & 17 & : \quad 1 \\
 & \downarrow \times 5 & \downarrow \times 5 \\
 & 85 \text{ wicket} & 5 \text{ Wicket}
 \end{array}$$

The number of wickets taken by him till the last match was $\Rightarrow 85 + 5 = \boxed{90} \text{ Ans}$

110 The average score of a cricketer for 20 matches is 52 runs. His highest score is more than its lowest score by 120 runs. If these two innings are excluded, the average of the remaining 18 matches is 50 runs. The highest score of the player is:

Soln:- Total score of 20 matches = $20 \times 52 = 1040$

Total score of 18 matches = $18 \times 50 = 900$

max + min. score = $1040 - 900 = 140$

Given max - min = 120

max = $\frac{120+140}{2} = \frac{260}{2} = \boxed{130}$ Ans

111 The average age of a cricket team of 11 players is the same as it was 3 years back because 3 of the players whose current average age of 33 years are replaced by 3 youngsters. The average age of the new comers is:

Soln:- Increased age of 11 players = $11 \times 3 = 33$ years

current age of 3 players who are replaced

= $3 \times 33 = 99$

Age of 3 New comers = $99 - 33 = 66$ years

avg = $\frac{66}{3}$

= $\boxed{22 \text{ years}}$ Ans

112 The average marks of Ravi in five subjects are 150, but in mathematics 43 was misread as 23 during the calculation. The correct average is:

Soln:- New increased Avg = $150 + \frac{(43-23)}{5}$
 $= 150 + 4 = \boxed{154}$ Ans

113 In an exam, the average marks obtained by the students was found to be 60. After omission of computational errors, the average marks of 100 candidates had to be changed from 60 to 30 and the average with respect to all the examinees came down to 45 marks. The total number of candidates who took the exam, was:

Soln:- let the number of students = x

$$\frac{60x - (60 \times 100) + (30 \times 100)}{x} = 45$$

$$60x - 3000 = 45x$$

$$x = \boxed{200}$$
 Ans

114 Several students have taken an exam. There was an error in the answer key which affected the marks of 48 students, and their average marks reduced from 78 to 66. The average of remaining students increased by 3.5 marks. This resulted the reduction of the average of all students by 4.5 marks. The number of students that attended the exam is:

Solⁿ:- Alligation:-

$$(78-66)$$

$$\downarrow$$

$$-12 \quad +3.5$$

$$-4.5$$

$$8 : 7.5$$

$$16 : 15$$

$$\times 3 \quad \times 3$$

$$48 : 45$$

students whose attend the exam,

$$\Rightarrow \boxed{93}$$

Ans

115 The number of students in section A and section B of a class are 40 and 52, respectively. The average score in mathematics of all the students is 75. If the average score of the students in A is 20% more, than of students in B, then what is the average score of students in B?

Solⁿ:-

Avg of A

Avg of B

$$6x$$

$$5x$$

$$40 \times 6x + 52 \times 5x = (40+52) \times 75$$

$$240x + 260x = 92 \times 75$$

$$500x = 92 \times 75$$

$$x = 13.8$$

$$\text{avg of B} = 5x$$

$$= 5 \times 13.8$$

$$= \boxed{69} \text{ Ans}$$

116 The number of students in a class is 75, out of which $33\frac{1}{3}\%$ are boys and the rest are girls. The average score in mathematics of the boys is $66\frac{2}{3}\%$ more than that of the girls. If the average score of all the students is 66, then the average score of the girls is:

Soln:-

	B	G
	1	: 2
avg →	<u>5x</u>	<u>3x</u>

$$33\frac{1}{3}\% = \frac{1}{3}$$

$$66\frac{2}{3}\% = \frac{2}{3}$$

$$5x + 2 \times 3x = 3 \times 66$$

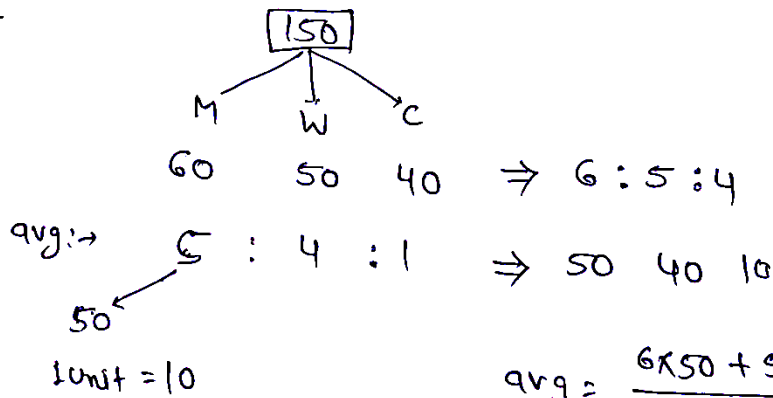
$$11x = 3 \times 66$$

$$x = 18$$

$$\text{avg of girls} = 3x = 3 \times 18 = \boxed{54} \text{ Ans}$$

117 In a group of 150 people, $\frac{2}{5}$ are men, $\frac{1}{3}$ are women and the rest are children. The average age of the women is $\frac{4}{5}$ of the average age of the men. The average age of the men. The children is $\frac{1}{5}$ of the average age of the men. If the average age of the men is 50 years, then the average age of all the people in the group is:

Soln:-



$$\text{avg} = \frac{6 \times 50 + 5 \times 40 + 40}{15}$$

$$= \frac{300 + 200 + 40}{15} = \boxed{36 \text{ years}}$$

118 A librarian purchased 50 story books for his library. But he saw that he could get 14 books more by spending ₹76 more but per book average becomes ₹1 less. The average price of each book he bought was.

Soln:- Let the average price of 1 book = x

$$\frac{50x + 76}{(50+14)} = (x-1)$$

$$50x + 76 = 64(x-1)$$

$$14x = 140$$

$$\boxed{x = 10} \text{ Ans}$$

119 The average of n numbers is 45. If 60% of the numbers are increased by 5 each and the remaining numbers are decreased by 10 each, then what is the average of the number so obtained?

Soln:- Let $n=100$

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$$\text{New avg} = \frac{100 \times 45 + 60 \times 5 - 40 \times 10}{100} \Rightarrow \boxed{44} \text{ Ans}$$

120 A man spends his three months income in four month time. If his monthly income is ₹1,000 then his annual saving is:

Soln:- Annual income = $1000 \times 12 = 12000$

Annual Expenditure = $1000 \times 9 = 9000$

Saving = $12000 - 9000$

= $\boxed{3000} \text{ Ans}$